

Flexible Demand Coordination & Interoperability

Meeting with CEC Interoperability Working Group

DC Jackson

Two Grid Coordination Problems

"Macro" Grid Coordination

Managing the Grid

- Highly-dynamic prices
- Demand-response events
- Functional control of devices
- Grid events (FlexAlerts)

"Micro" Grid Coordination

Managing the Distribution Network

- Per-transformer capacity management
- Dynamic control of customer site import & export
- Can't do much of this today — static sizing based on predicted worst-case load

Analytical Framework

Explicit vs. Implicit Demand Response

- **Legacy DR:** Explicit events — shed, load-up, un-shed
- **Dynamic pricing:** Implicit DR — device behavior driven by cost + customer preference

| *Price is superior to explicit DR events*

The Role of Functional Control

- Ultimately, device behavior is altered via functional-control commands
- Optimal translation of grid signals to device commands requires **local context**
- The further from the customer this translation occurs, the less context is available
- The customer **must** choose where translation occurs: device, local EMS, or aggregator
- Translation **must not** be locked inside the manufacturer's cloud

The Grid Should Manage the Grid, Not Devices

- Functional control of individual devices at the grid level was once necessary — it is now **obsolete**
- Grid, LSEs, and aggregators should coordinate via **objectives** (prices, DR events), not device control
- Price alone is **necessary but not sufficient** for distribution management
- Explicit power limits are also required at the customer-site level

Capacity Management

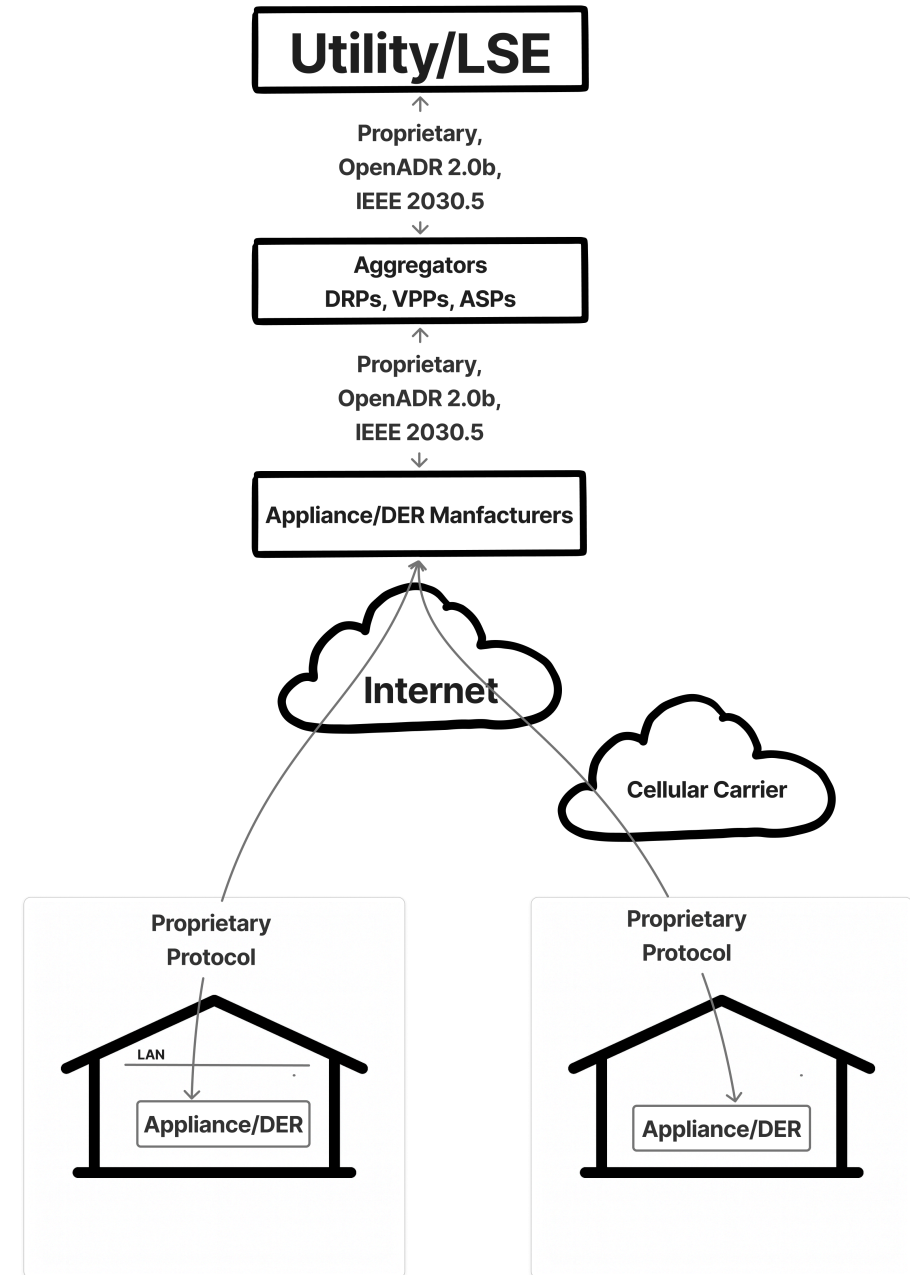
- Smart panels now enable dynamic per-site power control (NEC-705.13 & UL-3141)
- Management should be at the **customer site**, not individual devices
- Each site may have: PV, BESS, EVSE, V2X-EVSE, HVAC, water heater, etc.

How Is Residential Coordination Done Today?

The Aggregator Model

Each appliance connects to its **manufacturer's cloud** via proprietary protocol. Aggregators contract with manufacturers; utilities contract with aggregators.

No one has a holistic view of the customer's energy situation.



Problems with Proprietary Protocols

- Customer cannot centrally manage devices from different manufacturers
- No direct customer access to power/energy management APIs
- No local control without cloud/Internet
- Customer usage data sent to cloud, likely sold to third parties
- Manufacturer cloud discontinuation breaks grid coordination
 - *Google recently halted smart features of Gen 1 & 2 Nest thermostats*
- Use of an aggregator should be customer's **choice**, not a requirement
- Customers have a direct relationship with the utility — they **must** have the option to communicate directly
- Proprietary protocols rarely support price response

Principles for Flexible Demand Interoperability

1. An **open protocol** must be supported for all grid-coordination features
2. The appliance must **directly implement** the protocol
3. The protocol must operate over **IP networks**
4. The appliance must support Ethernet, Wi-Fi, and/or cellular network interfaces
5. The customer must be able to **configure the server** with which the appliance communicates
6. Configuration must **not require** the manufacturer's app or cloud
7. The protocol must support **local or cloud** communication, whichever the customer chooses
8. The protocol must **not require an aggregator**
9. Utilities must support **direct customer connection** to their grid-coordination servers using an open protocol

Protocol Comparison

Criterion	IEEE 2030.5	OCPP	CTA-2045	AHRI 1380	OpenADR 3.1	Matter
Open standard	✓	✓	✓	✓	✓	✓
Dynamic pricing	✓				✓	✓
DR events	✓	✓	✓	✓	✓	✓
Local control	✓	✓			✓	✓
Cloud-to-appliance					✓	
Customer-configurable server					✓	
Appliance-integrated			✓		✓	✓
Site power management					✓	✓
All appliance types	✓				✓	✓

Why Not IEEE 2030.5?

- Not used by LSEs or aggregators to directly control residential BTM DERs
- Manufacturers either translate 2030.5 to proprietary commands, or proxy requests
- No LSE support for direct customer connection over the Internet
- No flexible-demand appliances support 2030.5 directly
- Customers cannot configure 2030.5 DER clients
- No coordination with customer's local EMS

Conclusion: *2030.5 is not a viable candidate for residential flexible demand*

Why Not CTA-2045?

- Only offered on water heaters — estimated 2-4% nationwide have CTA-2045 ports
- Requires aftermarket UCM dongle (~\$100 in volume)
- UCM-to-aggregator protocol is proprietary and non-standard
- Doesn't support dynamic prices
- In practice: proven insurmountable deployment obstacle

The assumptions are obsolete

- Appliances **do** contain microcontrollers now
- Network interface costs **are** low enough for integration

Conclusion: *CTA-2045 has failed to achieve material deployment. CEC FDAS should not specify it.*

Why Not AHRI 1380?

- HVAC systems supporting AHRI 1380-2019 connect to the **manufacturer's cloud** via proprietary protocol
- Manufacturer's cloud then provides OpenADR 2.0b and/or CTA-2045-A to DR providers
- The HVAC appliance itself offers **no open protocol** for demand flexibility

Does not support:

- Direct appliance support for an open protocol
- Connection to a customer's local EMS
- Customer choice of grid-coordination server

Conclusion: AHRI 1380-2019 does not meet interoperability principles. CEC FDAS should not specify it.

Why OpenADR 3.1?

OpenADR 3.1 can do everything 2.0b does (with less cost and complexity), plus:

- **Cloud-to-cloud, cloud-to-appliance, and local-EMS-to-appliance** communication
- Modern web technologies: HTTPS/TLS, JSON, OAuth2, MQTT
- Implemented on **inexpensive microcontrollers** (ESP32, <\$5 in volume)
- Supports most grid coordination functions for flexible demand

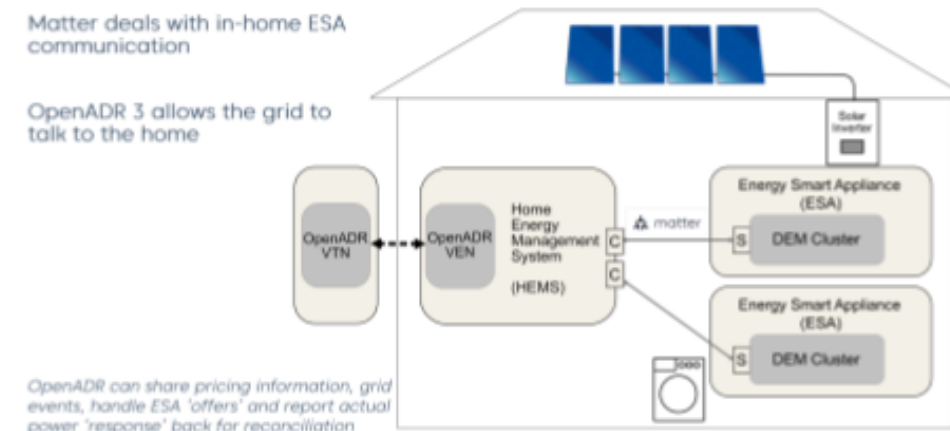
Conclusion: *OpenADR 3.1 should be the first-choice protocol for flexible demand appliances, DERs, and LSE/aggregator servers*

Matter: The Local Complement

- Matter excels at **smart-appliance interoperability within the home**
- Does **not** provide utility-to-customer coordination alone
- Combined with OpenADR 3.1: EMS receives grid signals via OpenADR, controls appliances via Matter
- Matter Device Energy Management (DEM) clusters enable standardized local control

OpenADR 3.1 + Matter = Complete Solution

- OpenADR 3.1 for **grid-to-home** coordination (prices, DR events, power limits)
- Matter for **within-home** device control and energy management
- Customer's EMS bridges between the two protocols



Proposed CEC FDAS Mandates

OpenADR 3.1+ must be supported by all flexible-demand system actors

LSEs must:

- Provide OpenADR 3.1+ VTNs supporting Internet connections from customers and aggregators

Flexible demand appliances must support:

- OpenADR 3.1+ over IP networks
- Customer-configurable VTN server address
- Network interfaces: Ethernet, Wi-Fi, and/or cellular

Flexible demand appliances should support:

- Matter Device Energy Management (DEM) clusters
- Appliance-specific Matter clusters (EVSE Mode, Water Heater, Thermostat)

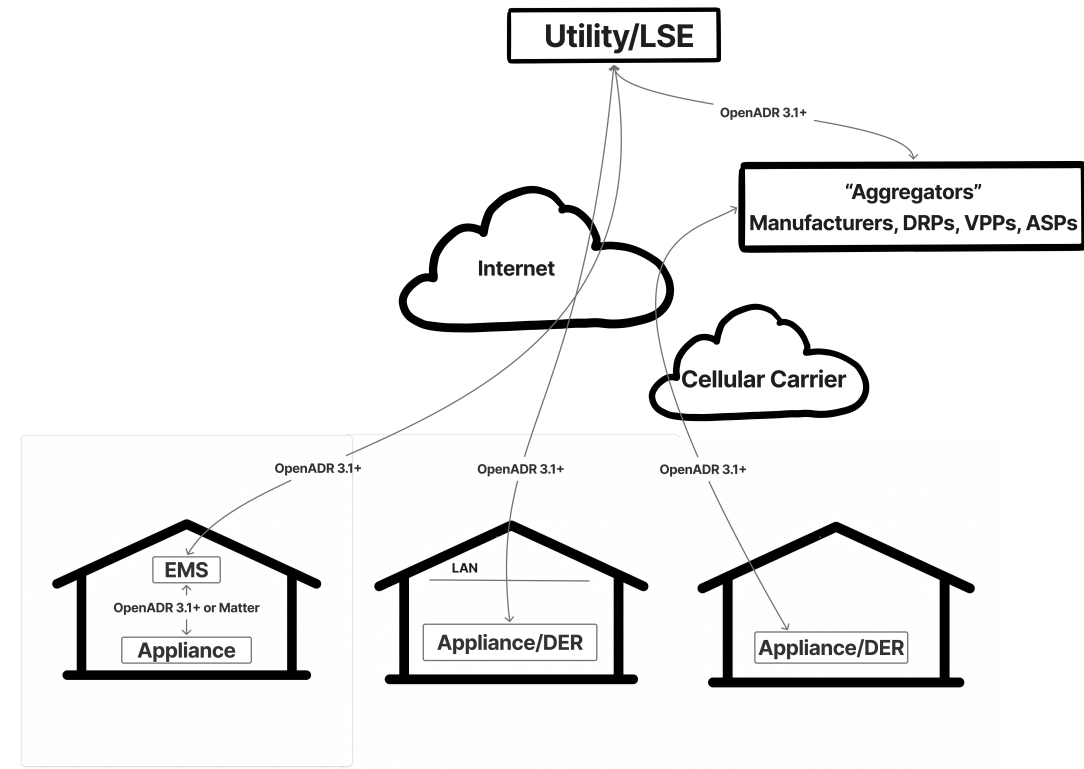
The Proposed System

How it works:

1. Appliance comes pre-configured with manufacturer's VTN URL
2. On connection, searches for local EMS via mDNS
3. If discovered, connects to local EMS instead
4. Customer can reconfigure VTN URL at any time: to LSE, aggregator, or local EMS

All parties coordinate via one open protocol: OpenADR 3.1+

- Customer connects to LSE, any aggregator, or local EMS by changing one URL
- Aggregators, ASPs, manufacturers, utilities all speak the same protocol



Future Directions

- Standardization of general device functional control is challenging
- IEEE 2030.5 provides strong control for **inverters only**
- OpenADR 3 currently provides rudimentary device control
- **Matter is making excellent progress** on standardized functional control

The path forward:

- Local EMS receives grid coordination via OpenADR 3
- Translates to functional control via Matter
- LSE "inverter control" must transition from **individual inverter** to **customer site**
- Enhance OpenADR 3 to deliver CSIP profiles for multi-DER sites

The convergence of OpenADR 3 and Matter enables the open, interoperable, customer-centric grid coordination architecture.